UNIVERSITY NAME

Analysis V — Final Exam

Year: 2023

Exercise 1:

Let $E = c^2([0,1], \mathbb{R})$. For every $f \in E$, define:

$$N(f) = \int_0^1 |f(z)|\, dz, \quad N'(f) = |f(0)| + \int_0^1 |f'(z)|\, dz, \quad N''(f) = |f(0)| + |f'(0)| + \int_0^1 |f''(z)|\, dz.$$

(a) Show that N, N', N'' are norms on E.

(b) Prove that for every $f \in E$, $N(f) \leq N'(f) \leq N''(f)$.

(c) Show that these norms are pairwise inequivalent.

2. Consider the map $d : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^+$ defined by:

$$d(x, y) = \frac{\|x - y\|}{1 + \|x - y\|}$$

where $\| \cdot \|$ is the usual Euclidean norm on $\mathbb{R}^n$. Show that d is a distance on $\mathbb{R}^n$.

Answer Area

Part 1

(a) Norm verification

A norm must satisfy: positivity, homogeneity, and triangle inequality.

- **Positivity**: For each N, N', N'', the expression is ≥ 0 as sum/integral of absolute values. If the norm equals 0:
  - For N: $\int_0^1 |f| = 0 \Rightarrow f \equiv 0$ (by continuity).
  - For N': $|f(0)| = 0$ and $\int_0^1 |f'| = 0 \Rightarrow f' \equiv 0$ so f constant; with $f(0) = 0 \Rightarrow f \equiv 0$.
  - For N'': similar reasoning gives $f \equiv 0$.
- **Homogeneity**: Clear since $|\lambda f| = |\lambda||f|$, same for derivatives.
- **Triangle inequality**: Follows from triangle inequality for absolute value and integrals.

Thus N, N', N'' are norms.

(b) Inequalities

- $N(f) \le N'(f)$: For any $x \in [0,1]$,

$$|f(x)| = \left| f(0) + \int_0^x f'(t)dt \right| \le |f(0)| + \int_0^1 |f'(t)|dt.$$

Integrate over $[0,1]$:

$$\int_0^1 |f(x)|dx \le |f(0)| + \int_0^1 |f'(t)|dt = N'(f).$$

- $N'(f) \le N''(f)$: Similarly,

$$|f'(x)| = \left| f'(0) + \int_0^x f''(t)dt \right| \le |f'(0)| + \int_0^1 |f''(t)|dt.$$

Integrate and add $|f(0)|$:

$$N'(f) \le |f(0)| + |f'(0)| + \int_0^1 |f''(t)|dt = N''(f).$$

(c) Non-equivalence

Two norms are equivalent if $\exists c, C > 0$ such that $c\|f\|_a \le \|f\|_b \le C\|f\|_a$ for all f.

- N and N': Take $f_n(x) = x^n$. Then

$$N(f_n) = \frac{1}{n+1} \to 0, \quad N'(f_n) = 0 + \int_0^1 nx^{n-1}dx = 1.$$

So $N(f_n)/N'(f_n) \to 0$, no lower bound.

- N' and N'': Take $f_n(x) = \sin(n\pi x)$. Then

$$N'(f_n) \sim n, \quad N''(f_n) \sim n^2 \quad \text{as } n \to \infty.$$

Ratio $N'(f_n)/N''(f_n) \to 0$.

- N and N'': Similarly not equivalent.

Part 2: Distance verification

Define $d(x,y) = \frac{\|x-y\|}{1+\|x-y\|}$ with $\|\cdot\|$ Euclidean.

1. **Positivity**: $d(x,y) \ge 0$, and $d(x,y) = 0 \Leftrightarrow \|x-y\| = 0 \Leftrightarrow x = y$.
2. **Symmetry**: $d(x,y) = d(y,x)$ since $\|x-y\| = \|y-x\|$.
3. **Triangle inequality**: Let $\phi(t) = \frac{t}{1+t}$ for $t \ge 0$. ϕ is increasing and concave. For $a = \|x-y\|, b = \|y-z\|$:

$$d(x,z) = \phi(\|x-z\|) \le \phi(a+b) \le \phi(a) + \phi(b) = d(x,y) + d(y,z),$$

where the last inequality holds because:

$$\frac{a+b}{1+a+b} \le \frac{a}{1+a} + \frac{b}{1+b}.$$

(Verified by cross-multiplying: $ab(2+a+b) \ge 0$.)

Thus d is a distance.

Exercise 2:

(1) Let $f : (E, \|\cdot\|_E) \to (F, \|\cdot\|_F)$ be a map. Show that the following properties are equivalent:

(a) f is continuous;
(b) For every $A \subset E$, $f(\overline{A}) \subset \overline{f(A)}$;
(c) For every $B \subset F$, $f^{-1}(\overline{B}) \supset \overline{f^{-1}(B)}$;
(d) For every $B \subset F$, $f^{-1}(\overline{B}) \subset \overline{f^{-1}(B)}$.

(2) Let K be a compact subset of a finite-dimensional normed vector space E. Show that there exists an open set U in E such that U is compact and $K \subset U$.

Answer Area

Part 1: Equivalent characterizations of continuity

We show $(a) \Leftrightarrow (b) \Leftrightarrow (c) \Leftrightarrow (d)$.
(a) ⇒ (b)

Assume f continuous. Let $A \subset E$, $y \in f(\overline{A})$. Then $y = f(x)$ with $x \in \overline{A}$. So $\exists (x_n) \subset A$ such that $x_n \to x$. By continuity, $f(x_n) \to f(x) = y$. Thus $y \in \overline{f(A)}$. Hence $f(\overline{A}) \subset \overline{f(A)}$.
(b) ⇒ (c)

Assume (b). Let $B \subset F$, and set $A = f^{-1}(B)$. Then by (b):

$$f(\overline{f^{-1}(B)}) \subset \overline{f(f^{-1}(B))} \subset \overline{B}.$$

Thus if $x \in \overline{f^{-1}(B)}$, then $f(x) \in \overline{B}$, so $x \in f^{-1}(\overline{B})$. Hence $\overline{f^{-1}(B)} \subset f^{-1}(\overline{B})$.
(c) ⇒ (a)

Assume (c). Let $C \subset F$ be closed. Then $\overline{C} = C$. Using (c) with $B = C$:

$$\overline{f^{-1}(C)} \subset f^{-1}(\overline{C}) = f^{-1}(C).$$

But $f^{-1}(C) \subset \overline{f^{-1}(C)}$, so equality holds. Thus $f^{-1}(C)$ is closed. Therefore f is continuous.
(a) ⇒ (d)

Assume f continuous. Let $B \subset F$. Since $\overline{B}$ is closed, $f^{-1}(\overline{B})$ is closed (by continuity). Also $f^{-1}(B) \subset f^{-1}(\overline{B})$. The closure $\overline{f^{-1}(B)}$ is the smallest closed set containing $f^{-1}(B)$, so

$$\overline{f^{-1}(B)} \subset f^{-1}(\overline{B}).$$

(d) ⇒ (a)

Assume (d). Let $C \subset F$ be closed. Then $\overline{C} = C$. Using (d) with $B = C$:

$$f^{-1}(C) = f^{-1}(\overline{C}) \subset \overline{f^{-1}(C)}.$$

But $\overline{f^{-1}(C)} \subset f^{-1}(C)$ always (since closure is larger). Thus equality: $f^{-1}(C) = \overline{f^{-1}(C)}$, so $f^{-1}(C)$ is closed. Hence f continuous.

Thus all four properties are equivalent.

Part 2: Compact subset in finite-dimensional space

Let $K \subset E$ be compact. Since E is finite-dimensional, all norms are equivalent; use the Euclidean norm. Since K is compact, it is bounded: $\exists R > 0$ such that $K \subset \overline{B}(0, R)$ (closed ball). Define

$$U = B(0, R + 1)$$

the open ball of radius $R + 1$.

- U is open.
- $\overline{U} = \overline{B}(0, R + 1)$ is closed and bounded in finite dimensions, hence compact (Heine-Borel).
- $K \subset B(0, R) \subset B(0, R + 1) = U$.

Thus U is an open set with compact closure containing K.

Exercise 3:

(1) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \dfrac{xy(y^2 - x^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

(a) Study the continuity of f on $\mathbb{R}^2$.

(b) Compute the partial derivatives of f at $(0, 0)$.

(c) Is f differentiable at $(0, 0)$?

(d) Determine the partial derivatives $\frac{\partial f}{\partial x}(x, y)$ and $\frac{\partial f}{\partial y}(x, y)$ on $\mathbb{R}^2 \setminus \{(0, 0)\}$.

(e) Show that f is of class C^1 on $\mathbb{R}^2$.

(f) Is f of class C^2 on $\mathbb{R}^2$?

(2) Consider the equation (E) : $x^3 + y^3 - x^2y - 1 = 0$ in $\mathbb{R}^2$.

(a) Show that equation (E) defines implicitly a function $y = \phi(x)$ in a neighborhood of 0 such that $\phi(0) = 1$.

(b) Show that ϕ has a local minimum at 0.

(c) Give the equation of the tangent line to the curve $y = \phi(x)$ at 0.

Answer Area

Part 1

(a) Continuity

For $(x, y) \neq (0, 0)$, f is rational hence continuous. At $(0, 0)$, use polar coordinates: $x = r\cos\theta$, $y = r\sin\theta$:

$$f(r\cos\theta, r\sin\theta) = \frac{r^2\cos\theta\sin\theta \cdot r^2(\sin^2\theta - \cos^2\theta)}{r^2} = r^2\cos\theta\sin\theta(\sin^2\theta - \cos^2\theta).$$

Since $|\cos\theta\sin\theta(\sin^2\theta - \cos^2\theta)| \leq 1$, we have $|f| \leq r^2 \to 0$ as $r \to 0$. Thus

$$\lim_{(x,y)\to(0,0)} f(x, y) = 0 = f(0, 0),$$

so f is continuous on $\mathbb{R}^2$.

(b) Partial derivatives at $(0, 0)$

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h\to 0}\frac{f(h,0) - f(0,0)}{h} = \lim_{h\to 0}\frac{0}{h} = 0.$$
$$\frac{\partial f}{\partial y}(0,0) = \lim_{k\to 0}\frac{f(0,k) - f(0,0)}{k} = \lim_{k\to 0}\frac{0}{k} = 0.$$

(c) Differentiability at $(0, 0)$

Check the limit:

$$\lim_{(h,k)\to(0,0)} \frac{f(h,k) - f(0,0) - 0\cdot h - 0\cdot k}{\sqrt{h^2+k^2}} = \lim_{(h,k)\to(0,0)} \frac{f(h,k)}{\sqrt{h^2+k^2}}.$$

In polar: $\frac{r^2\cos\theta\sin\theta(\sin^2\theta - \cos^2\theta)}{r} = r\cos\theta\sin\theta(\sin^2\theta - \cos^2\theta) \to 0$ as $r \to 0$ (bounded times r). Thus f is differentiable at $(0, 0)$ with differential 0.

(d) Partial derivatives for $(x, y) \neq (0, 0)$

Compute via quotient rule. Write $f = \frac{xy^3 - x^3y}{x^2+y^2}$.

$$\frac{\partial f}{\partial x} = \frac{(y^3 - 3x^2y)(x^2+y^2) - (xy^3 - x^3y)(2x)}{(x^2+y^2)^2} = \frac{y(y^4 - x^4 - 4x^2y^2)}{(x^2+y^2)^2}.$$

$$\frac{\partial f}{\partial y} = \frac{(3xy^2 - x^3)(x^2+y^2) - (xy^3 - x^3y)(2y)}{(x^2+y^2)^2} = \frac{x(y^4 - x^4 + 4x^2y^2)}{(x^2+y^2)^2}.$$

(e) Class C^1

We already have partials everywhere. Check continuity at $(0, 0)$: In polar,

$$\left|\frac{\partial f}{\partial x}\right| \leq \frac{r \cdot r^4}{r^4} = r \to 0,$$

so $\frac{\partial f}{\partial x}$ continuous at $(0, 0)$. Similarly for $\frac{\partial f}{\partial y}$. Thus $f \in C^1(\mathbb{R}^2)$.

(f) Class C^2?

Compute mixed partials at $(0,0)$:

$$\frac{\partial^2 f}{\partial y \partial x}(0,0) = \lim_{k\to 0} \frac{\frac{\partial f}{\partial x}(0,k) - \frac{\partial f}{\partial x}(0,0)}{k} = \lim_{k\to 0} \frac{k-0}{k} = 1.$$

$$\frac{\partial^2 f}{\partial x \partial y}(0,0) = \lim_{h\to 0} \frac{\frac{\partial f}{\partial y}(h,0) - \frac{\partial f}{\partial y}(0,0)}{h} = \lim_{h\to 0} \frac{-h-0}{h} = -1.$$

Since $\frac{\partial^2 f}{\partial y\partial x}(0,0) \neq \frac{\partial^2 f}{\partial x\partial y}(0,0)$, by Clairaut's theorem f cannot be C^2 at $(0,0)$. So $f \notin C^2(\mathbb{R}^2)$.

Part 2: Implicit function

(a) Existence of ϕ

Let $F(x,y) = x^3 + y^3 - x^2y - 1$. Then $F(0,1) = 0$. Compute:

$$\frac{\partial F}{\partial y}(x,y) = 3y^2 - x^2, \quad \frac{\partial F}{\partial y}(0,1) = 3 \neq 0.$$

By the Implicit Function Theorem, there exists a C^1 function ϕ defined near $x = 0$ with $\phi(0) = 1$ such that $F(x, \phi(x)) = 0$.

(b) Local minimum at 0

Differentiate $F(x, \phi(x)) = 0$:

$$3x^2 + 3\phi(x)^2\phi'(x) - 2x\phi(x) - x^2\phi'(x) = 0.$$

At $x = 0$, $\phi(0) = 1$: $0 + 3\phi'(0) - 0 - 0 = 0 \Rightarrow \phi'(0) = 0$.
Differentiate again:

$$6x + 6\phi(x)\phi'(x)^2 + 3\phi(x)^2\phi''(x) - 2\phi(x) - 2x\phi'(x) - 2x\phi'(x) - x^2\phi''(x) = 0.$$

At $x = 0$, $\phi(0) = 1$, $\phi'(0) = 0$: $0 + 0 + 3\phi''(0) - 2 - 0 - 0 - 0 = 0 \Rightarrow \phi''(0) = \frac{2}{3} > 0$.
Thus ϕ has a local minimum at $x = 0$.

(c) Tangent line

Slope $\phi'(0) = 0$, point $(0,1)$. Equation: $y = 1$.